Food Analytical Methods (2025) 18:385–397
https://doi.org/10.1007/s12161-024-02728-0

RESEARCH

Qualitative Identification and Adulteration Quantification of Extra Virgin Olive Oil Based on Raman Spectroscopy Combined with Multi-task Deep Learning Model

Shuxin Liang[1,2] · Guoqing Chen[1,2] · Chaoqun Ma[1,2] · Jiao Gu[1,2] · Chun Zhu[1,2] · Lei Li[1,2] · Hui Gao[1,2] · Zichen Yang[1,2] · Jun Cao[1,2] · Zehao Chen[1,2]

Received: 20 April 2024 / Accepted: 21 November 2024 / Published online: 27 November 2024
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024

Abstract
An extra virgin olive oil (EVOO) adulteration detection method based on Raman spectroscopy and a single-model multi-task deep learning network (MTDL) model is proposed to simultaneously achieve qualitative identification and quantitative analysis of olive oil blends. Soybean oil, peanut oil, sunflower oil, corn oil, and palm oil were blended into extra virgin olive oil at different concentrations, and a total of 675 spectra were collected for five samples with different olive oil contents. Analysis and visualization of spectral datasets are provided using dimensionality reduction algorithms. The data enhancement technique resulted in a classification accuracy of 99.3% for the qualitative analysis of the MTDL and a good linear fit for the concentration dataset of different types of samples, with RMSEP and *R*-squared reaching 6.0910 and 0.9909, respectively. Compared to other classification algorithms (PLS-DA, SVM, *k*-NN, random forest) and regression methods (PLSR, SVR), MTDL exhibits notable performance and efficiency, showing potential for simultaneously conducting qualitative identification and quantitative analysis of olive oil products with Raman spectroscopy.

Keywords Raman spectroscopy · Multi-task deep learning · Extra virgin olive oil · 1D CNN · Classification · Adulteration quantification

Introduction

With the continuous improvement of living standard, people pay more and more attention to safety and health in diet. Olive oil, as a healthy cooking oil, is highly favored by consumers. Its richness in antioxidants, healthy fats, and anti-inflammatory properties has made it an indispensable ingredient in many households and restaurants. However, as the olive oil market continues to expand and the complexity of the global supply chain increases, the quality and purity of olive oil products is becoming an issue of great concern (Zarezadeh et al. 2021).

According to the International Olive Council classification scale, olive oil can be categorized into several types, including extra virgin olive oil, virgin olive oil, olive oil, and blended olive oil (Bongartz and Oberg 2011). EVOO is considered to be the highest quality olive oil and is usually obtained by mechanical cold pressing of olives and has a very low acidity, usually below 0.8%. And because no chemicals or heat treatments are added, EVOO has the purest flavor and aroma, which results in a higher price. In addition, in the vegetable blending oil market, EVOO is often used as a premium fat in vegetable blending oils due to its richness in monounsaturated fatty acids (Zhu et al. 2023a), which have the effect of improving cardiovascular health and lowering cholesterol levels. Some companies are involved in blending low-quality vegetable oils into olive oils while falsely labeling the olive oil content of their products in order to reduce production costs and make more profit (Vali Zade et al. 2023). Therefore, the rapid and efficient identification and quantification of EVOO content are of paramount importance for safeguarding consumer rights and ensuring effective market regulation.

In past studies, chemical analytical methods, such as gas chromatography (Purcaro et al. 2016) and liquid

✉ Guoqing Chen
jncgq@jiangnan.edu.cn

1 School of Science, Jiangnan University, Wuxi, China

2 Jiangsu Provincial Research Center of Light Industrial Optoelectronic Engineering and Technology, Wuxi, China

chromatography (Hidalgo-Ruiz et al. 2019), were usually used to identify the components in edible oils. These methods often require complex reagents and equipment, are cumbersome and time-consuming to operate, and are more destructive to the samples. Raman spectroscopy, as a non-invasive, rapid, and highly sensitive analytical method that provides information on the molecular structure of the sample and good discrimination between different kinds of compounds, has received increasing attention in the field of olive oil detection (Hidalgo-Ruiz et al. 2019; Duraipandian et al. 2019).

Deep learning can be used to process large amounts of unstructured data, which has been applied to a variety of fields such as image recognition (Xiao et al. 2020) and machine vision (Huang et al. 2017). Deep learning algorithms are applied to spectral analysis techniques by constructing a one-dimensional convolutional neural network (1D CNN). A large amount of spectral data is used for training to develop qualitative and quantitative models for high accuracy classification and quantitative recognition tasks (Lu et al. 2023; Luo et al. 2022; Zeng et al. 2023). In recent years, detection methods based on Raman spectroscopy and deep learning have gradually become the focus of researchers. The prediction of olive oil content in mixed corn-olive oils was realized using 1D CNN networks combined with Raman spectroscopy (Wu et al. 2022a). As well as recognizing three kinds of mixed olive oils with 100% accuracy (Wu et al. 2022b). In practical field detection, the strategy of qualitative classification followed by quantitative prediction is generally used (Wu et al. 2023, 2022c). However, for detection scenarios containing multiple adulterated substances, a large number of models need to be pre-built and trained to cope with different combinations of adulteration. Therefore, constructing a single-model multi-task deep learning architecture to learn and train multiple related tasks simultaneously can improve efficiency and obtain better generalization ability and performance than single-task models in practical detection applications (Xie et al. 2022).

In this study, an EVOO adulteration detection method combining Raman spectroscopy and a multi-task deep learning model is developed. A single-model multi-task architecture was used to extract features from spectral data, and the classification task and concentration prediction task were trained simultaneously for spectral identification and quantitative analysis. Based on 675 Raman spectra of five vegetable oils with different olive oil contents, the spectral dataset was visualized using principal component analysis (PCA) and t-distributed stochastic nearest neighbor embedding (t-SNE). After extending the dataset by data enhancement techniques, the multi-task deep learning model achieved the desired results. Compared with SVM, PLS-DA, *k*-NN, and random forest, the multi-task deep learning model achieves 99.3% classification accuracy and better linear fit than PLSR and SVR with RMSEP = 6.0910 and *R*-squared = 0.9909. It is expected that in the future, this method could be further extended to address more specific tasks and applied to portable Raman spectrometers, providing a solution for large-scale, multi-label oil detection and analysis.

Materials and Methods

Sample Preparation

We purchased refined palm oil (Flower Core brand, Thailand), peanut oil, corn germ oil, soybean oil, sunflower oil (Goldfish brand, China), and extra virgin olive oil from three different brands (BORGES, YBARRA, Oleo Bella brands, Spain) from Tmall. Then, samples of palm extra virgin olive blend oil (PO-EVOO), peanut extra virgin olive blend oil (PN-EVOO), corn extra virgin olive blend oil (CO-EVOO), soybean extra virgin olive blend oil (SB-EVOO), and sunflower extra virgin olive blend oil (SF-EVOO) were made. The olive oils were blended with each of the refined oil samples at concentration ratios of 10% (v/v), 20% (v/v), 30% (v/v), 40% (v/v), 50% (v/v), 60% (v/v), 70% (v/v), 80% (v/v), and 90% (v/v), respectively. To simulate adulteration, nine concentration ratios were established. A comprehensive set of 135 samples was meticulously prepared, taking into account three distinct brands of EVOO and five types of adulterants (refined oils). The selection of samples was based on the consideration of both the type of olive oil and the corresponding adulterant.

Taking SF-EVOO sample with 30% (v/v) olive oil as an example, use a pipette to transfer 7 ml of sunflower oil and 3 ml of olive oil into a clean, dry centrifuge tube. Then, stir the mixture three times using an ultrasonic oscillator, each time for three minutes. The process for creating samples with other ratios is the same. All experiments are conducted at room temperature.

Spectra Acquisition and Preprocessing

Spectra were collected using confocal Raman spectrometer (Renishaw, England) with a laser source of 532 nm, laser power of 10%, integration time of 10 s, grating of 1800 g/mm, and a lens with a 50 × objective. The spectral acquisition range was set to 800–3100 cm^{-1} based on the commonly studied bands of edible oils. Measurements were made at five different randomly selected locations on the samples, yielding 675 spectra for 135 samples.

The Raman spectra obtained from the instrument typically encompass signals unrelated to the sample information, including fluorescence background, light scattering effects, and instrumental noise. Therefore, eliminating irrelevant information by preprocessing the spectral data

can speed up the training of the model and make the loss curve converge faster literature. The collected data are preprocessed as follows:

1. **Smoothing.** Savitsky-Golay smoothing is used to eliminate random noise from the original Raman spectral data.
2. **Scattering correction.** Standard normal variable transform (SNV) is used to correct the data for light scattering, which can effectively eliminate the spectral differences due to different scattering levels, thus enhancing the correlation between the spectra and the data.
3. **Baseline correction.** Adaptive iterative reweighted penalized least squares (airPLS) is used to remove baseline drift in Raman spectral data due to a variety of factors (Zhang et al. 2010), such as instrumental noise, sample inhomogeneity, solvent absorption, and scattering.
4. **Normalization.** The maximum-minimum normalization method (Xu et al. 2020) is used to process the data and map the data between [0, 1] to ensure that the data are comparable in the same dimensions so that the network model can converge faster and better.

Data Augmentation

It is worth noting that for deep learning networks, a large number of dataset samples are required for training. Due to the highly sensitive nature of Raman spectroscopy, the prediction results of the model may be influenced by variations in the intensity of Raman characteristic peaks for the same sample across different measurements, arising from changes in laser power, variations in sampling position, and fluctuations in excitation efficiency (Zhu et al. 2023b). Therefore, several data enhancement methods with Gaussian noise superimposed were applied to generate training samples with diversity to simulate spectral variations under different conditions according to the possible transformations of the data during the acquisition process (Xie et al. 2022). We increased the total number of spectra from 675 to 3240 for MTDL model training. The methods are summarized as follows: (i) random mixing, (ii) amplitude warping, (iii) axial warping, and (iv) shifting. The effect of data augmentation methods is shown in Fig. S1.

Convolutional Neural Network

The overall architecture of the single-model multi-task CNN network is shown in Fig. 1. We conducted multiple rounds of training and experimentation on our dataset to

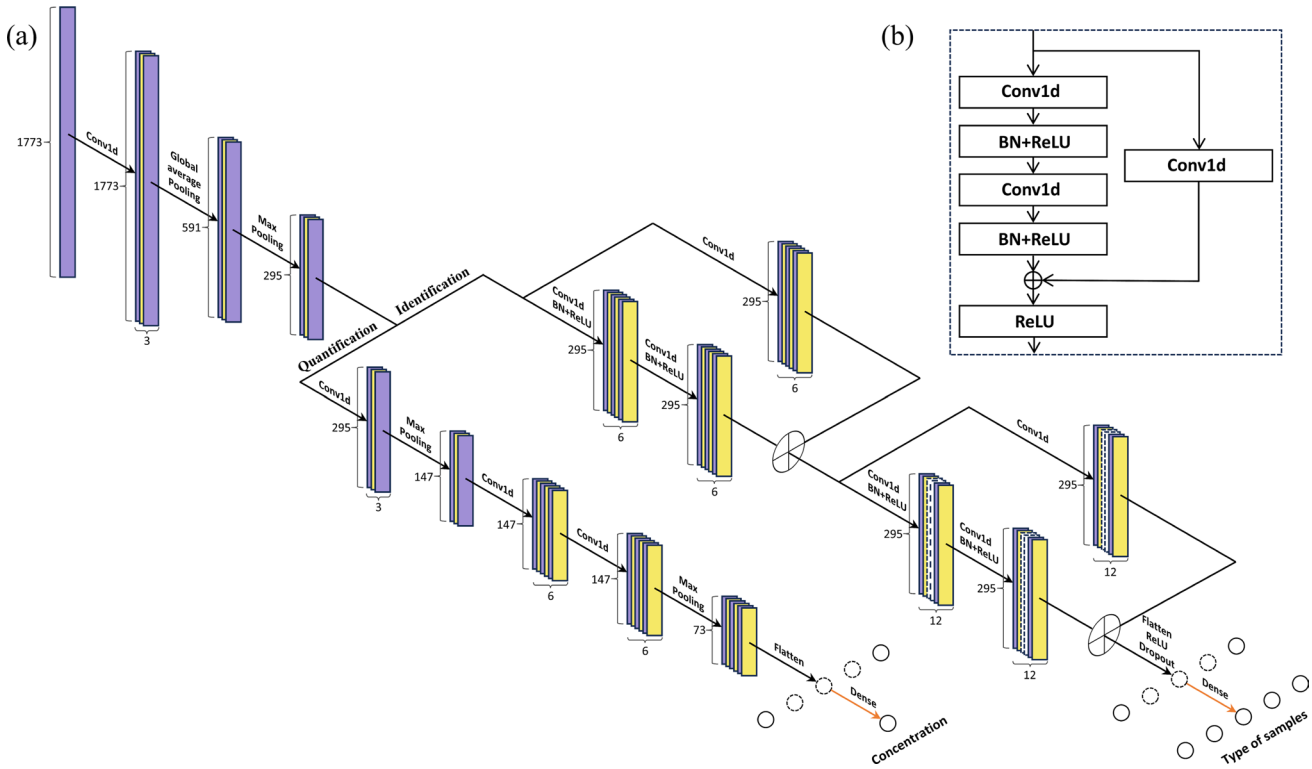

Fig. 1 **a** The structure of multitask deep learning model based on 1D CNN; **b** Residual module

arrive at the current multi-task structure. The model consists of a shared data feature extraction layer at the bottom and two task modules. Spectral data is first input to the qualitative and quantitative task modules through a convolutional layer, a global average pooling layer, and a maximum pooling layer. As the model deepens, the gradual increase in channels enhances its ability to capture subtle and complex patterns within the spectral data, enabling more refined feature extraction at each layer. ResNet was first proposed by He et al. in 2015 and has shown significant advantages in classification tasks (He et al. 2016). In the qualitative analysis for mixed olive oil identification, two residual modules were used for feature extraction to enhance the model's classification performance. The abstract features obtained from these residual modules are converted into one-dimensional vectors using a flatten layer, and then output as 5 output values through a dense layer. The values are transformed into the predicted probabilities of the five blended oils using the SoftMax function, and the label corresponding to the maximum probability is taken, which is the predicted sample type. Figure 1b illustrates a residual module comprising a shortcut connection and a main path. The main path consists of multiple nonlinear transformations, with each convolutional layer followed by Batch Normalization (BN) and the nonlinear activation function ReLU, to stabilize the training process and accelerate convergence. This design enables the network to efficiently learn the residuals between the outputs and inputs of the primary path and can greatly facilitate the optimization process of deep networks (Ioffe and Szegedy 2015). For the task of olive oil concentration prediction, we drew on the structure of previous quantitative CNN models (Wu et al. 2022a, c). Following bottom feature extraction, the data were input to a quantitative model consisting of three convolutional layers and two maximum pooling layers after bottom feature extraction, and finally outputted as a predicted value through the flatten and dense layers. Detailed information of the whole model can be found in Table S1.

In model training, the loss function is crucial for adjusting weights and parameters to bring predictions closer to actual targets (Xie et al. 2022). Different loss functions are taken for various tasks. Cross-entropy loss (CE) measures the discrepancy between predicted probability distributions and true labels in qualitative classification tasks, while mean squared error (MSE) loss, typical in regression problems, calculates the average squared differences between predicted and actual values in quantitative tasks. Different from the typical single-task model, we combine the weighted individual task loss functions as the total loss function for the multi-task model.

A learning rate decay method is introduced to prevent the model from exploding the gradient due to excessive learning rate during training, and the current learning rate is multiplied by 0.4 every 50 epochs. The initial learning rate is set to 0.005, the batch size is set to 64, the dropout is set to 0.1, and the Adam optimizer is used. After 200 iterations, the learning rate converges to 0 and training is stopped.

Chemometrics and Machine Learning Algorithms

Spectral datasets were visualized and analyzed using PCA and t-SNE. PCA is able to project the high-dimensional data into a low-dimensional subspace to maximize the preservation of the variance of the original data. It is often used as a dimensionality reduction method in food Raman fingerprint analysis (Morais et al. 2019). Based on the characterization of spectral intensity values, t-SNE use the t-distribution to describe the relationship between the spectra, mapping the spectral data to a lower dimensional space by minimizing the difference in similarity between the high and low dimensional spaces. t-SNE not only takes into account the linear relationships between data points, but is also able to capture complex nonlinear structures in the data. This makes it more powerful in dealing with high-dimensional data, while it is often used in data visualization to discover the clustering structure and inter-cluster relationships of data (Chakravartula et al. 2022).

k-NN is a supervised learning method which is distance-based and evaluates the similarity between different training instances in the feature space by measuring their distances from each other (Reokrungruang et al. 2019). One of the significant advantages of *k*-NN is the simplicity and rapidity of the learning process, since the model does not need to learn the parameters explicitly. This makes *k*-NN an intuitive and easy-to-understand algorithm that is particularly suitable for the case of small datasets.

As a commonly used machine learning model in spectral detection, random forest is an integrated learning approach that builds on multiple decision trees. Random forest constructs multiple different decision trees by randomly sampling training data and features, and integrates their predictions by voting or averaging to improve the generalization ability of the overall model (Seifert 2020).

Support vector machine (SVM) is a typical machine learning algorithm. The core concept is to separate the dataset into different classes by finding an optimal hyperplane and maximizing the interval between the classes. SVM have excellent performance in high-dimensional spaces and in contexts with small sample data (Yang et al. 2023). When dealing with continuous olive oil concentration labels, SVR can be implemented to fit the data and predict the output of continuous values by minimizing the prediction error, called SVR. Radial basis function (RBF) is chosen as the kernel function for SVM and SVR because of its good fitting performance and adaptive ability to deal with nonlinear relationships between data well.

Partial least squares regression (PLSR) and partial least squares discriminant analysis (PLS-DA) are multivariate statistical learning methods based on the least squares method, which is widely used to deal with high-dimensional data and to solve covariance problems. PLSR searches for latent variables (LVs) by minimizing the sum of squares of residuals between the inputs and outputs in order to establish a relationship (Medina et al. 2019). PLS-DA is a classification extension of PLSR that focuses on efficiently dealing with classification problems by introducing category information in modeling and projecting multivariate data into a latent variable space to maximize the differences between categories (Pomerantsev and Rodionova 2018). Both methods perform well in the analysis of small samples of high-dimensional data and provide powerful tools for prediction and classification tasks in the field of spectral detection.

Results and Discussion

Raman Spectral Analysis

Figure 2a displays the average Raman spectra of various types of edible oils. It can be noticed that the pretreatment Raman spectra of edible oils are very similar in terms of peak positions and shapes, and most of them only reflect differences in intensity, which may be due to the fact that edible oils contain similar main components. And the differences between different oils are more reflected in the content of various fatty acids. Fatty acids usually consist of long carbon chains, which usually have between 12 and 22 carbon atoms, and can be categorized into saturated and unsaturated fatty acids (Zhu et al. 2023a). The characteristic peaks at 1154 cm^{-1} and 1525 cm^{-1} are related to the carotenoids that are characteristic of virgin olive oil, and originate from the C–C and C = C stretching vibrations, respectively, in highly conjugated compounds such as carotenoids (El-Abassy et al. 2009). Carotenoids are gradually destroyed as the refining process of edible oils intensifies, so they are only present in EVOO. The adulteration of poor

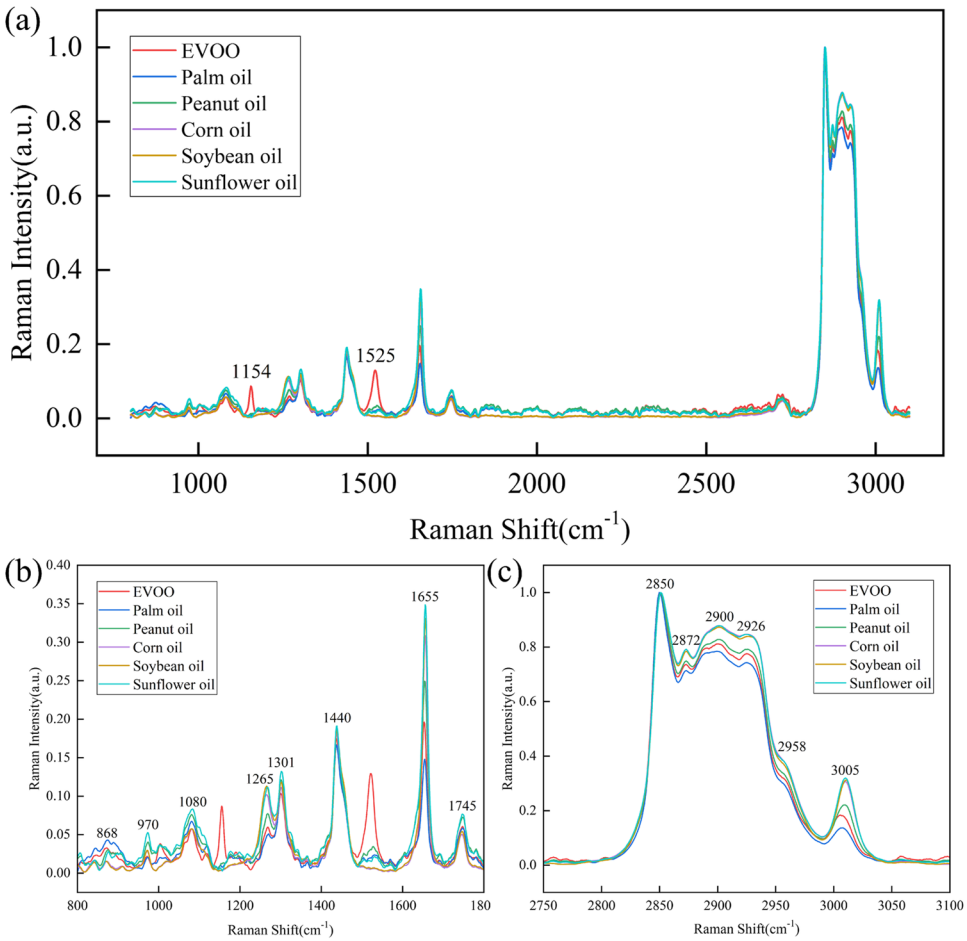

Fig. 2 **a** Average Raman spectra of different edible oils. **a** Full spectra; **b** spectra of 800–1800 cm^{-1}; **c** spectra of 2750–3100 $cm.^{-1}$

quality or refined oils reduces the carotenoid content of olive oil, so the carotenoid content is commonly used to identify adulteration of olive oil (Muik et al. 2005). The Raman spectra of edible oils were studied mainly in the region of 800–1800 and 2750–3100 cm^{-1} wave numbers (Carmona et al. 2014; Jiménez-Sanchidrián and Ruiz 2016), as shown in Fig. 2b and c. In the low wavelength region, the characteristic peaks at 868, 1080 cm^{-1} wave number are related to the C–C stretching vibration in fatty acids, reflecting the degree of unsaturation of edible oils. 970, 1655 cm^{-1} peaks are attributed to the bending vibration of the trans(C = C) and the cis(C = C) stretching vibration in unsaturated fatty acids, respectively. 1265 cm^{-1} is the bending vibration peak of C-H in olefin (RHC = CHR). 1301 cm^{-1} and 1440 cm^{-1} represent the curling vibration and shearing vibration of CH_2, which are mainly related to the saturated functional group. 1745 cm^{-1} is the stretching vibration peak of the C = O in ester bond (Zhu et al. 2023a; Wu et al. 2022b). In the wave number range of 2700–3100 cm^{-1}, the strong peak at 2850 cm^{-1} and the small peak at 2872 cm^{-1} correspond to the symmetric stretching vibrations of CH_2 and CH_3, respectively. The scattering located near 2900 cm^{-1} is attributed to the antisymmetric stretching vibration of CH_2 in the long carbon chain that is distant from the carbon–carbon double bond as well as from the methyl group. The shoulder peak at 2926 cm^{-1} corresponds to the antisymmetric stretching of CH_2 adjacent to methyl groups. The peak at 2958 cm^{-1} is due to the simplex stretching of CH_3, while the 3005 cm^{-1} peak is attributed to the symmetric stretching of C-H near C = C (Carmona et al. 2014).

The signals of olive, palm, peanut, and sunflower oils are more ambiguous in the signal-free band, primarily due to their strong fluorescent background. This results in higher noise levels post-de-baselining, consequently lowering the signal-to-noise ratio.

Taking SF-EVOO as an example, Fig. 3 illustrates the average Raman spectra of oils with varying adulteration ratios. It is evident that the spectra of samples with varying extra virgin olive oil (EVOO) ratios exhibit a high degree of similarity. Notably, a discernible trend is observed in the attenuation or enhancement at specific wavenumbers, namely 1154, 1265, 1525, 1655, 2900, and 3005 cm^{-1}. The two enhanced peak spectra (1154 and 1525 cm^{-1}) and three weakened peaks (1265, 1655, 3005 cm^{-1}) with increasing olive oil concentration were attributed to changes in the carotenoid and fatty acid content of the system, respectively. EVOO contains 11% saturated fatty acids and about 80% monounsaturated fatty acids, and its oleic acid ($C_{18}H_{34}O_2$) content is twice that of other oils. Sunflower oil mainly consists of the polyunsaturated fatty acid, linoleic acid ($C_{18}H_{32}O_2$), accounting for about 65% and featuring more C = C double bonds. Therefore, as the concentration of olive oil in the mixture increased, the peak intensities at 1265, 1655, and 3005 cm^{-1} showed a decreasing trend. The intensity change near 2900 cm^{-1} was related to the fact that sunflower oil contains more long carbon chain fatty acids.

In addition, certain remaining peaks do not exhibit direct intensity differences related to concentration, but the chemical bonding and molecular structure information they contain can also indirectly indicate changes in the samples at the microscopic level related to adulteration (Rodionova et al. 2021). We can notice that the addition of adulterants in olive oil affects the overall trend of the Raman spectral curve. This suggests that Raman spectroscopy, being a versatile analytical testing method, is well-suited for comprehensively characterizing chemical bonding changes in edible oil samples, making it a viable tool for detecting olive oil fraud.

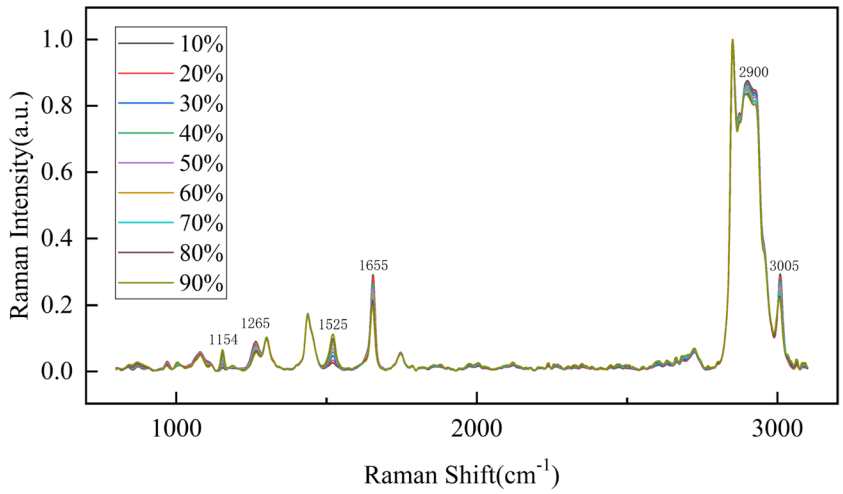

Fig. 3 Preprocessed Raman spectra of EVOO adulterated with sunflower oil at different concentrations

Visualization and Analysis of Spectral Dataset

In order to gain a deeper understanding of the structure and characteristics of the dataset, we first employed PCA for data visualization and dimensionality reduction. Through PCA, we successfully revealed the distribution locations and clustering trends of different types of olive oil adulteration samples in the fraction space in Fig. 4a. The three largest principal components of the PCA (variance contribution values PC1: 59.11%, PC2: 17.93%, and PC3: 8.23%) depicted 85.27% of the cumulative variance contribution. For the multiple blended oil adulteration dataset, the different samples showed a banded distribution in space along different directions and a tendency to cluster between classes. CO-EVOO, SB-EVOO, and SF-EVOO samples are not well differentiated, especially when there is a serious overlap between the low concentration adulterated samples, which confirms the similarity of the three oils in terms of compositional composition (Teng et al. 2024). Figure 4b shows a plot of PCA scores for the dataset labeled with different olive oil contents. In Fig. 4b, the data show more of the same characteristics as the olive oil content increases, with the mixed samples gradually formed tight clusters in the negative direction of the PC2 axis. PC2 providing the main observation on variation in olive oil concentration in the dataset than the other principal components. In contrast, in the low olive oil concentration samples, the data show more single oil characteristics with a larger scatter distribution, thus differentiating between different types of oil mixtures mainly based on the PC1 and PC3.

In the case of adulteration of multiple refined oils, PCA does not allow a better visualization of the spectral dataset.

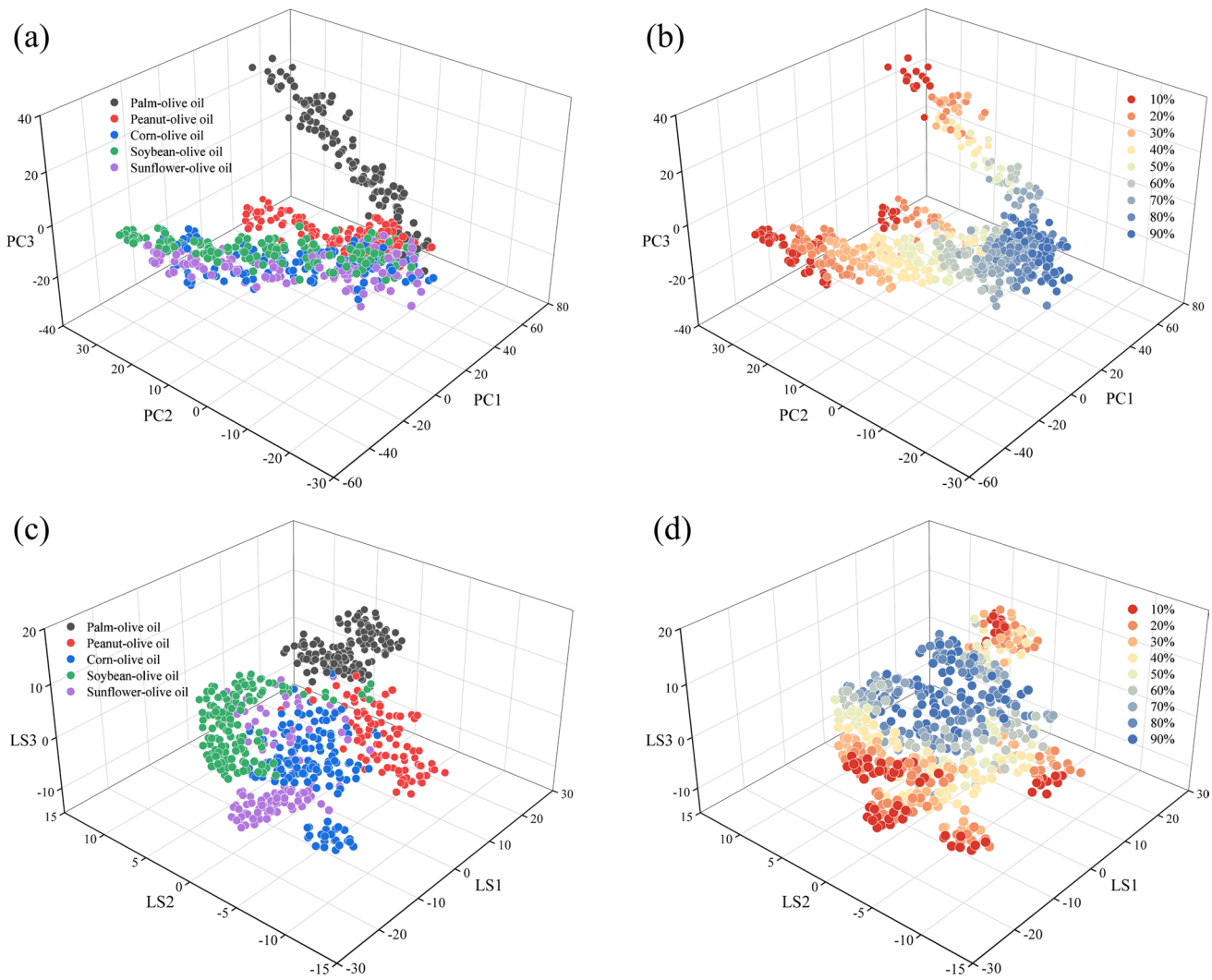

Fig. 4 PCA score plots of Raman spectra dataset for five adulterated EVOO with **a** different categories and **b** different EVOO concentrations. The spatial distribution plots of spectral dataset for five adulterated EVOO based on t-SNE with **c** different categories and **d** different EVOO concentrations in three-dimensional latent spaces

Therefore, a nonlinear dimensionality reduction technique, t-SNE, was used for the analysis of the spectral dataset. In contrast to PCA, which focuses on linear relationships between data points, t-SNE considers both linear and complex nonlinear structures in the data. This makes t-SNE a potent method for handling high-dimensional data while also improving data visualization performance. In the same case of the five distinct adulteration percentage sample datasets, t-SNE provides visual representations of the oil category in Fig. 4c and the adulteration concentrations in Fig. 4d.

In the latent space, despite the slight overlap of distribution points in the low concentrations, a clear separation trend of different oil categories can still be observed. Specifically, it was able to further distinguish olive oil samples adulterated with corn oil, soybean oil, and sunflower oil adulterants at low mixing concentrations (10%, 20%, and 30%). However, samples with higher olive oil concentrations (80% and 90%) still exhibited some degree of confusion.

Overall, the visualization of Raman spectra achieved by t-SNE is better than PCA, which provides a reference for subsequent qualitative and quantitative analysis.

Multi-task Deep Learning Model

The variability of the blended oil samples during spectral analysis and dataset visualization provides potential information and guidance for subsequent modeling. We utilize these differences as input features to apply MTDL model or other relevant algorithms for further classification or regression analysis.

In the training process of the model, the dataset was divided into training, validation, and test sets for training and testing of the deep learning model. For each of the 15 spectra of each concentration sample of each type of blended oil, three combinations of each are randomly selected as the test set, and the rest are merged into the initial training and validation dataset for further data enhancement process. Of the expanded 3240 data samples, 80% is used as the training set for training the deep learning model, while 20% of the data is used as the validation set for evaluating the model fit on the training set to adjust the model hyperparameters.

Figure 5 shows the loss curves of the MTDL model on the training and validation sets, which is one of the crucial metrics for evaluating the model's performance. We can see that in the early stages of training phase, the loss curve decreases more rapidly, showing a better convergence rate. The model then falls into a local optimum solution at a higher initial learning rate. Near about the 50th iteration, the loss curves of the training and validation sets decreased rapidly and converged to 0. Subsequently, the loss curves remained stable without major fluctuations. This indicates that the model converges and has a stable fit. The outputs of the model for 10 consecutive training sessions are shown in Table 1.

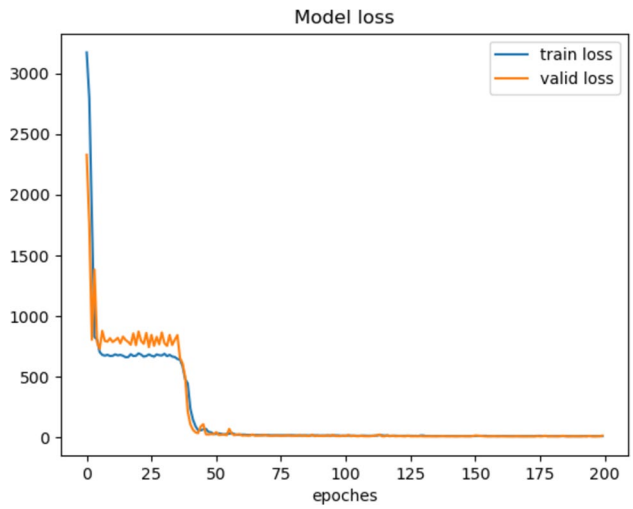

Fig. 5 Variation curves of loss value in MTDL model

Table 1 Results of 10 consecutive runs of MTDL model

Number	Accuracy (5 categories)	R_P^2	RMSEP
1	99.3%	0.9878	8.1485
2	99.3%	0.9891	7.2830
3	97.8%	0.9886	7.6131
4	98.5%	0.9903	6.4524
5	99.3%	0.9858	9.4941
6	98.5%	0.9870	8.6917
7	99.3%	0.9908	6.1432
8	98.6%	0.9909	6.0910
9	98.5%	0.9892	7.1967
10	99.3%	0.9865	8.9699
Average	**98.8%**	**0.9886**	**7.6083**

Qualitative Recognition of Adulterated Oil Matrices

The classification accuracy of the multitask deep learning model was 98.8% for 10 successive averages. Accurate classification of five mixed olive oil spectra was achieved by plotting the confusion matrix (Fig. S2) at best results and calculating precision, recall, f1-score, and accuracy.

In machine learning, spectral data is randomly partitioned, with 80% used for model training and 20% as a separate test set. Tenfold cross-validation is used to evaluate the performance of the model so that the model can utilize the data more efficiently with a smaller dataset. This approach reduces the bias in model performance evaluation due to improper data partitioning and provides more reliable performance estimates.

Figure 6a, b, c, and d demonstrate the PLS-DA, SVM, *k*-NN, and random forest classification result plots. When the

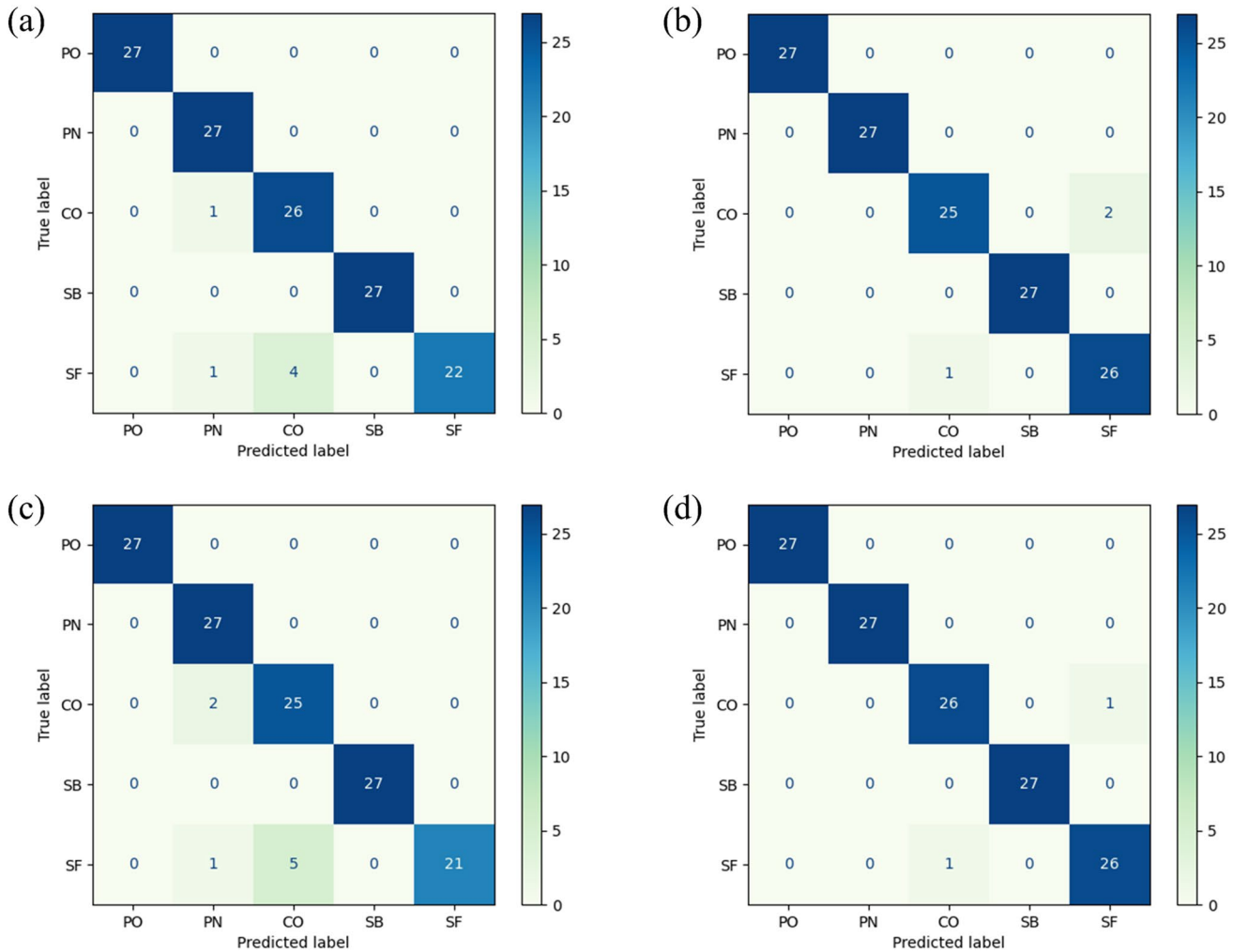

Fig. 6 Confusion matrix classification results obtained from **a** PLS-DA, **b** SVM, **c** *k*-NN, and **d** random forest

number of LVs is set to 9, the PLS-DA model achieves the best prediction performance on the test set with an accuracy of 94.1%. As well as the optimal accuracy of 97.8% for SVM when the regularization parameter $c = 10$. These machine learning algorithms succeeded in making a clear distinction for PO-EVOO, PN-EVOO, and SB-EVOO, but it is difficult to achieve complete separation for CO-EVOO and SF-EVOO. In particular, *k*-NN accurately predicted only 77% of SF-EVOOs in the test set, while 14.8% of SF-EVOOs were misclassified as CO-EVOO samples by random forests and 3.7% of CO-EVOOs and SF-EVOOs were misclassified as PN-EVOO samples by PLS-DA. This should also confirm that due to the very high similarity of the Raman spectra of maize and sunflower seeds, there is an overlap in their PCA analysis maps. The classification accuracy of PLS-DA with *k*-NN can reach 94.1%. Random forest classification exhibits slightly lower performance than MTDL, with only two samples misclassified out of 135 test samples. In contrast, MTDL constructed using 1D CNN achieves a unique and excellent performance while maintaining a stable and high classification accuracy. The accuracy results of the different classification models are shown in Table S2.

Quantitative Identification of EVOO in Vegetable Adulterated Oil

The predictions of PLSR, SVR, and MTDL for the detection of olive oil adulteration are presented in Fig. 7. The solid blue line represents the ideal regression line where the predicted value of adulteration concentration is equal to the true value, and the dashed line represents the fitted line with an absolute error of ± 4%.

In Fig. 7c, the MTDL model demonstrated the highest prediction accuracy with a coefficient of determination (R_p^2) of 0.9909 and a root mean square error percentage (RMSEP) of 6.0910. This performance reflects excellent agreement between predicted and actual values, and the residual distribution displayed minimal variability.

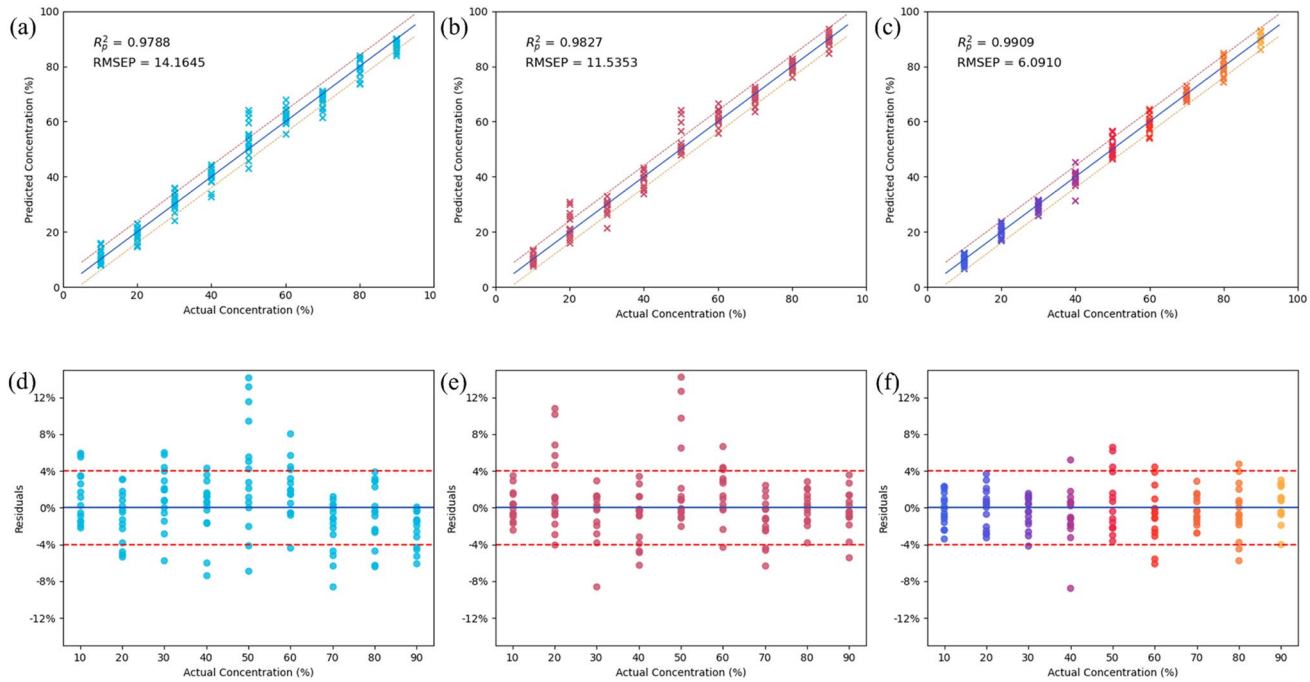

Fig. 7 Quantitative result and residual scatter plot of PLSR (**a**, **d**), SVR (**b**, **e**), and MTDL model (**c**, **f**)

In contrast, the PLSR model showed a slight lack of fit and achieved the best prediction accuracy on the test set with 9 LVs. In Fig. 7a, with an R_p^2 of 0.9788, the RMSEP increases to 14.1645, indicating a decrease in prediction performance, especially in the middle and high concentration regions where the prediction error is relatively elevated. For the concentration dataset containing multiple oils, the predicted values fitted with PLSR have a good linear relationship from an overall point of view (Fig. 7d). This may be due to the fact that PLSR is a combination of downscaling and regression analysis, which reduces the complexity of the data in dealing with high-dimensional data with multicollinearity (Medina et al. 2019).

The SVR model exhibited intermediate performance between PLSR and MTDL, achieving an R^2 of 0.9827 and an RMSEP of 11.5353. Its residuals displayed a uniform distribution, closely resembling the distribution of actual EVOO concentrations in Fig. 7e, signifying a strong generalization ability of the model. Nevertheless, there is still room for improvement in prediction accuracy, particularly in the 20% and 50% data bands.

The performance of three different regression models in predicting the olive oil concentration of the sample species was analyzed comparatively. As shown in Fig. 8, it is evident that the MTDL model exhibits the highest prediction accuracy, with a $R^2 = 0.9909$ and RMSEP = 6.0910. With excellent agreement between predicted and actual values, the residual distribution of MTDL model is showing low variability.

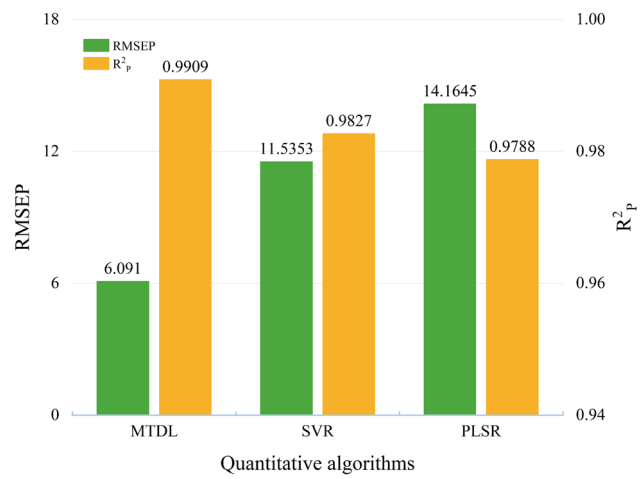

Fig. 8 Comparison of quantitative performance of different algorithms

To further validate the effectiveness of the proposed method, five bottles of EVOO products, including pure EVOO, vegetable blend oil with EVOO, and vegetable blend oil with olive oil, were purchased from the market. Detailed information is shown in Table S3. Table S4 shows all five products and their labeled concentrations, MTDL model result. The prediction accuracy of the model showed a decreasing trend with decreasing EVOO content and increasing composition. At the same time, the model has good specificity for the detection of extra virgin olive oil products and

can be free from the interference of other grades of olive oil. Although a two-component sample data was used during the training of the MTDL model, the classification results were able to represent the main constituent oils contained in EVOO product. For EVOO products labeled as 100% (P4, P5), the MTDL model predicted values exceeding the labeled amounts. This discrepancy may be attributed to the fact that the model training data used fewer brands of EVOO (3 brands) and different varieties of olives in the commercially available products.

In summary, we developed PLSR, SVR, and MTDL prediction models to analyze each olive oil sample, respectively. For the prediction of EVOO concentration, the MTDL model significantly outperformed PLSR in both chemometrics and SVR in machine learning. PLSR is limited by its linear approach in analyzing nonlinear spectral datasets, and it does not seem to take into account that the order of the wave numbers in the spectra is meaningful. Although the predictions of SVR are superior to those of PLSR, there is still room for improvement in its quantitative performance. Given the model complexity and computational cost considerations, the SVR model can also serve as a valid alternative. In contrast, as the principal element in the MTDL framework, CNN effectively extracts latent features from Raman spectral data (Debus et al. 2021). This feature extraction capability improves the efficiency of data processing and analysis, resulting in promising performance in both classification tasks and regression predictions in this study.

However, this work has certain areas for further exploration. First, the scale and diversity of the dataset may affect the model's generalizability, as the current dataset is based solely on binary EVOO mixtures. For applications involving multi-component mixtures or analyses of other types of oils, future efforts could introduce a more diverse dataset and make suitable adjustments to the multi-task model structure to enhance adaptability in more complex scenarios. In terms of model design, the flexibility of the multi-task learning architecture allows for the selection of different network structures tailored to specific tasks. Although the single-model multi-task framework offers some efficiency advantages in the training phase compared to multiple independent single-task models, its higher complexity may increase hardware requirements. Future research could explore more efficient implementations or simplify the model structure to improve its practical applicability. In addition, the MTDL model in this study is built on a 1D CNN that automatically captures hierarchical and subtle patterns within the data, making it well-suited for complex tasks. This automated feature extraction minimizes the extensive preprocessing steps typically required by traditional methods. Moreover, deep learning models like CNN have better transferability, as they can adapt learned features to new datasets or tasks with minimal retraining. Other architectures, such as recurrent neural network (RNN) and Transformer models, which excel at extracting temporal sequence information, could be explored in future studies to further improve predictive performance.

Conclusion

In this study, a new method for EVOO adulteration detection is proposed. We designed a multi-task deep learning network model based on Raman spectroscopy that can be used to simultaneously enable qualitative identification and quantitative analysis of EVOO. Spectra containing samples of five adulterated olive oils in different concentrations were collected. PCA and t-SNE effectively demonstrated the major structures and associations in the entire spectral dataset, successfully providing a clear presentation of the variability between samples. By grouping all the measured spectral data into one dataset and optimizing the hyperparameters such as the loss function, the model achieves a classification accuracy of up to 99.3% and a good linear fit with RMSEP = 6.0910 and $R^2 = 0.9909$ for concentration prediction. Compared to traditional classification algorithms, SVM, random forest, and MTDL have achieved excellent performance in olive oil blend recognition. In terms of quantitative analysis of concentration datasets containing five oils, MTDL performed better than PLSR and SVR models.

Overall, the identification and quantification results obtained by the MTDL model are impressive, with practical applicability for routine detection of EVOO products. The model structure that integrates qualitative recognition and quantitative analysis has the potential for extension by adding more task branches to perform additional detection tasks simultaneously. This strategy could improve the detection efficiency while reducing the training time and maintaining high performance. Therefore, the spectral analysis model proposed in this work provides a convenient and efficient approach for EVOO quality monitoring, demonstrating potential for large-scale, multi-label Raman spectral detection.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s12161-024-02728-0.

Author Contribution S.X.L. conceived and designed the study and wrote the initial draft. G.Q.C. managed the project and secured funding. C.Q.M. and Z.C.Y. reviewed and edited the manuscript. C.Z. and H.G. supplied resources. L.L. and H.G. supervised the project. Z.H.C. and J.C. conducted the research. All authors reviewed and approved the final manuscript.

Funding This work was financially supported by the National Natural Science Foundation of China (No. 62375112).

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

References

Bongartz A, Oberg DG (2011) Sensory evaluation of extra virgin olive oil (EVOO) extended to include the quality factor "harmony." J Agric Sci Technol 2011:422–435. https://doi.org/10.21256/zhaw-1864

Carmona MÁ, Lafont F, Jiménez-Sanchidrián C, Ruiz JR (2014) Raman spectroscopy study of edible oils and determination of the oxidative stability at frying temperatures. Eur J Lipid Sci Technol 116(11):1451–1456. https://doi.org/10.1002/ejlt.201400127

Chakravartula SSN, Moscetti R, Bedini G, Nardella M, Massantini R (2022) Use of convolutional neural network (CNN) combined with FT-NIR spectroscopy to predict food adulteration: a case study on coffee. Food Control 135:108816. https://doi.org/10.1016/j.foodcont.2022.108816

Debus B, Parastar H, Harrington P, Kirsanov D (2021) Deep learning in analytical chemistry. TrAC, Trends Anal Chem 145:116459. https://doi.org/10.1016/j.trac.2021.116459

Duraipandian S, Petersen JC, Lassen M (2019) Authenticity and concentration analysis of extra virgin olive oil using spontaneous Raman spectroscopy and multivariate data analysis. Appl Sci 9(12):2433. https://doi.org/10.3390/app9122433

El-Abassy RM, Donfack P, Materny A (2009) Rapid determination of free fatty acid in extra virgin olive oil by Raman spectroscopy and multivariate analysis. J Am Oil Chemists' Soc 86:507–511. https://doi.org/10.1007/s11746-009-1389-0

He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition 770–778. https://doi.org/10.48550/arXiv.1512.03385

Hidalgo-Ruiz JL, Romero-González R, Vidal JLM, Frenich AG (2019) A rapid method for the determination of mycotoxins in edible vegetable oils by ultra-high performance liquid chromatography-tandem mass spectrometry. Food Chem 288:22–28. https://doi.org/10.1016/j.foodchem.2019.03.003

Huang Z, Pan Z, Lei B (2017) Transfer learning with deep convolutional neural network for SAR target classification with limited labeled data. Remote Sensing 9(9):907. https://doi.org/10.3390/rs9090907

Ioffe S, Szegedy C (2015) Batch normalization: accelerating deep network training by reducing internal covariate shift. Proceedings of the 32nd International conference on machine learning. plmr 37:448–456.

Jiménez-Sanchidrián C, Ruiz JR (2016) Use of Raman spectroscopy for analyzing edible vegetable oils. Appl Spectrosc Rev 51(5):417–430. https://doi.org/10.1080/05704928.2016.1141292

Lu B, Tian F, Chen C, Wu W, Tian X, Chen C, Lv X (2023) Identification of Chinese red wine origins based on Raman spectroscopy and deep learning. Spectrochim Acta Part A Mol Biomol Spectrosc 291:122355. https://doi.org/10.1016/j.saa.2023.122355

Luo X, Gouda M, Perumal HZ, Lin L, Tang Y, Sanaeifar A, He Y, Li X, Dong C (2022) Using surface-enhanced Raman spectroscopy combined with chemometrics for black tea quality assessment during its fermentation process. Sens Actuators, B Chem 373:132680. https://doi.org/10.1016/j.snb.2022.132680

Medina S, Perestrelo R, Silva P, Pereira JA, Câmara JS (2019) Current trends and recent advances on food authenticity technologies and chemometric approaches. Trends Food Sci 85:163–176. https://doi.org/10.1016/j.tifs.2019.01.017

Morais CL, Paraskevaidi M, Cui L, Fullwood NJ, Isabelle M, Lima KM, Martin-Hirsch PL, Sreedhar H, Trevisan J, Walsh MJ (2019) Standardization of complex biologically derived spectrochemical datasets. Nat Protoc 14(5):1546–1577. https://doi.org/10.1038/s41598-020-62338-8

Muik B, Lendl B, Molina-Díaz A, Ayora-Cañada MJ (2005) Direct monitoring of lipid oxidation in edible oils by Fourier transform Raman spectroscopy. Chem Phys Lipid 134(2):173–182. https://doi.org/10.1016/j.chemphyslip.2005.01.003

Pomerantsev AL, Rodionova OY (2018) Multiclass partial least squares discriminant analysis: taking the right way-a critical tutorial. J Chemom 32(8):e3030. https://doi.org/10.1002/cem.3030

Purcaro G, Barp L, Beccaria M, Conte LS (2016) Characterisation of minor components in vegetable oil by comprehensive gas chromatography with dual detection. Food Chem 212:730–738. https://doi.org/10.1016/j.foodchem.2016.06.048

Reokrungruang P, Chatnuntawech I, Dharakul T, Bamrungsap S (2019) A simple paper-based surface enhanced Raman scattering (SERS) platform and magnetic separation for cancer screening. Sens Actuators, B Chem 285:462–469. https://doi.org/10.1016/j.snb.2019.01.090

Rodionova OY, Pierna JF, Baeten V, Pomerantsev A (2021) Chemometric non-targeted analysis for detection of soybean meal adulteration by near infrared spectroscopy. Food Control 119:107459. https://doi.org/10.1016/j.foodcont.2020.107459

Seifert S (2020) Application of random forest based approaches to surface-enhanced Raman scattering data. Sci Rep 10(1):5436. https://doi.org/10.1038/s41598-020-62338-8

Teng Y, Chen Y, Chen X, Zuo S, Li X, Pan Z, Shao K, Du J, Li Z (2024) Revealing the adulteration of sesame oil products by portable Raman spectrometer and 1D CNN vector regression: a comparative study with chemometrics and colorimetry. Food Chem 436:137694. https://doi.org/10.1016/j.foodchem.2023.137694

Vali Zade S, Forooghi E, Jannat B, Hashempour-baltork F, Abdollahi H (2023) A combined classification modeling strategy for detection and identification of extra virgin olive oil adulteration using Raman spectroscopy. Chemom Intell Lab Syst 240:104903. https://doi.org/10.1016/j.chemolab.2023.104903

Wu X, Gao S, Niu Y, Zhao Z, Ma R, Xu B, Liu H, Zhang Y (2022a) Quantitative analysis of blended corn-olive oil based on Raman spectroscopy and one-dimensional convolutional neural network. Food Chem 385:132655. https://doi.org/10.1016/j.foodchem.2022.132655

Wu X, Gao S, Niu Y, Zhao Z, Xu B, Ma R, Liu H, Zhang Y (2022b) Identification of olive oil in vegetable blend oil by one-dimensional convolutional neural network combined with Raman spectroscopy. J Food Compos Anal 108:104396. https://doi.org/10.1016/j.jfca.2022.104396

Wu X, Xu B, Ma R, Gao S, Niu Y, Zhang X, Du Z, Liu H, Zhang Y (2022c) Botanical origin identification and adulteration quantification of honey based on Raman spectroscopy combined with convolutional neural network. Vib Spectrosc 123:103439. https://doi.org/10.1016/j.vibspec.2022.103439

Wu X, Xu B, Luo H, Ma R, Du Z, Zhang X, Liu H, Zhang Y (2023) Adulteration quantification of cheap honey in high-quality Manuka honey by two-dimensional correlation spectroscopy combined with deep learning. Food Control 154:110010. https://doi.org/10.1016/j.foodcont.2023.110010

Xiao Q, Bai X, Gao P, He Y (2020) Application of convolutional neural network-based feature extraction and data fusion for geographical origin identification of radix astragali by visible/short-wave near-infrared and near infrared hyperspectral imaging. Sensors 20(17):4940. https://doi.org/10.3390/s20174940

Xie L, Shen Y, Zhang M, Zhong Y, Lu Y, Yang L, Li Z (2022) Single-model multi-tasks deep learning network for recognition

and quantitation of surface-enhanced Raman spectroscopy. Opt Express 30(23):41580–41589. https://doi.org/10.1364/OE.472726

Xu Y, Zhong P, Jiang A, Shen X, Li X, Xu Z, Shen Y, Sun Y, Lei H (2020) Raman spectroscopy coupled with chemometrics for food authentication: a review. TrAC, Trends Anal Chem 131:116017. https://doi.org/10.1016/j.trac.2020.116017

Yang Z, Chen G, Ma C, Gu J, Zhu C, Li L, Gao H (2023) Magnetic Fe_3O_4@ COF@ Ag SERS substrate combined with machine learning algorithms for detection of three quinolone antibiotics: ciprofloxacin, norfloxacin and levofloxacin. Talanta 263:124725. https://doi.org/10.1016/j.talanta.2023.124725

Zarezadeh MR, Aboonajmi M, Ghasemi Varnamkhasti M (2021) Fraud detection and quality assessment of olive oil using ultrasound. Food Sci Nutr 9(1):180–189. https://doi.org/10.1002/fsn3.1980

Zeng Y, Liu Z, Fan X, Wang X (2023) Modified denoising method of Raman spectra-based deep learning for Raman semi-quantitative analysis and imaging. Microchem J 191:108777. https://doi.org/10.1016/j.microc.2023.108777

Zhang ZM, Chen S, Liang YZ (2010) Baseline correction using adaptive iteratively reweighted penalized least squares. Analyst 135(5):1138–1146. https://doi.org/10.1039/B922045C

Zhu J, Rong Y, Jiang X, Qian H, Yu X, Chen Q (2023a) Raman spectroscopy coupled with metaheuristics-based variable selection models: a method for rapid determination of extra virgin olive oil content in vegetable blend oils. J Food Compos Anal 123:105503. https://doi.org/10.1016/j.jfca.2023.105503

Zhu J, Jiang X, Rong Y, Wei W, Wu S, Jiao T, Chen Q (2023b) Label-free detection of trace level zearalenone in corn oil by surface-enhanced Raman spectroscopy (SERS) coupled with deep learning models. Food Chem 414:135705. https://doi.org/10.1016/j.foodchem.2023.135705

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.